

Log Rank Tests

Log Rank Test

Compare survival between two groups (non-parametric).

Null Hypothesis

There is no difference in survival between the two groups

Alternative Hypothesis

There is a difference in survival between the two groups

Demo

Log Rank Test

Knowledge Check

True or False:

Based on the results of this test, we reject the null hypothesis at a five percent significance level.

```
Call:
survdifff(formula = Surv(mos_to_death, died) ~ female, data =
hrs_df)
```

n=2394, 23 observations deleted due to missingness.

	N	Observed	Expected	$(O-E)^2/E$	$(O-E)^2/V$
female=0	936	325	271	10.98	18.7
female=1	1458	354	408	7.27	18.7

Chisq= 18.7 on 1 degrees of freedom, p= 2e-05

**Respond on
Poll Everywhere**



<https://pollev.com/albertlee033>

Knowledge Check

True or False:

Based on the results of this test, we reject the null hypothesis at a five percent significance level.

```
Call:
survdifff(formula = Surv(mos_to_death, died) ~ female, data =
hrs_df)
```

n=2394, 23 observations deleted due to missingness.

	N	Observed	Expected	$(O-E)^2/E$	$(O-E)^2/V$
female=0	936	325	271	10.98	18.7
female=1	1458	354	408	7.27	18.7

Chisq= 18.7 on 1 degrees of freedom, **p= 2e-05**

**Respond on
Poll Everywhere**



<https://pollev.com/albertlee033>



University of California
San Francisco